

 **TARGET CAT 2025** 



MBA FASTRACK

QUANT : Number System

Factors &
Unit Digit Theorem

Lecture No.- 1

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Recap of Previous Lecture :

Arithmetic

TOPICS *to be covered*



- 1) Concepts & Numerical based on 'Number System Basics'
- 2) Concepts based on 'Unit Digit'
- 3) Concepts based on 'Factors'
- 4) Problems for Practice & CAT PYQs



Number System Basics :



Types of Numbers :

① Natural No.

1, 2, 3, 4, 5, ...

② Whole No.

0, 1, 2, 3, 4, 5, ...

③ Integers

... - 3, -2, -1, 0, 1, 2, 3, 4, ...

④ Even No.

2, 4, 6, 8, ...

⑤ Odd No.

1, 3, 5, 7, ...

⑥ Prime No.

2, 3, 5, 7, 11, 13, 17, 19, 23, ...

⑦ Composite No.

4, 6, 8, 9, 10, 12, 15, 16, ...



Number System Basics :



Co-prime

$1, 5 \leftarrow 1, 7$
 $(5, 7) \rightarrow \text{coprime} \checkmark$

$1, 7 \leftarrow 1, 3, 9$
 $(7, 9) \rightarrow \text{coprime} \checkmark$

$1, 2, 4 \leftarrow 1, 3, 9$
 $(4, 9) \rightarrow \text{coprime} \checkmark$

$1, 3, 9 \leftarrow 1, 3, 5, 15$
 $(9, 15) \rightarrow \text{not coprime} \times$

$1, 2, 4 \leftarrow 1, 2, 3, 6$
 $(4, 6) \rightarrow \text{not coprime} \times$

factorization

$$6 = 2 \times 3$$

$$8 = 2^3$$

$$12 = 2^2 \times 3$$

$$144 = 2^4 \times 3^2$$

2	144
2	72
2	36
2	18
3	9
3	3
	1

2	180
2	90
3	45
3	15
5	5
	1

$$180 = 2^2 \times 3^2 \times 5$$

factorials

$$3! \rightarrow 3 \times 2 \times 1$$

$$4! \rightarrow 4 \times 3 \times 2 \times 1$$

$$6! \rightarrow 6 \times 5 \times 4 \times 3 \times 2 \times 1$$



$$\frac{4}{8} = 0.5$$



Unit Digit Theorem :



<u>Remainder</u>	1	2	3	0
<u>Raise to the Power</u>	Raise to 1	Raise to 2	Raise to 3	Raise to 4
Unit Digit '0'	0	0	0	0
Unit Digit '1'	1	1	1	1
Unit Digit '2' →	2	4	8	6
Unit Digit '3' →	3	9	7	1
Unit Digit '4'	4	6	4	6
Unit Digit '5'	5	5	5	5
Unit Digit '6'	6	6	6	6
Unit Digit '7' →	7	9	3	1
Unit Digit '8' →	8	4	2	6
Unit Digit '9'	9	1	9	1

$$\left(\frac{3791}{8} \right) = 2$$

$$\left(\frac{29483574}{7} \right) = 9$$



Unit Digit Theorem :



<u>Remainder</u>	1	2	3 ↓	0
<u>Raise to the Power</u>	Raise to 1	Raise to 2	Raise to 3	Raise to 4
Unit Digit '0'	0	0	0	0
Unit Digit '1'	1	1	1	1
Unit Digit '2' →	2	4	8	6
Unit Digit '3'	3	9	7	1
Unit Digit '4'	4	6	4	6
Unit Digit '5'	5	5	5	5
Unit Digit '6' →	6	6	6	6
Unit Digit '7'	7	9	3	1
Unit Digit '8'	8	4	2	6
Unit Digit '9'	9	1	9	1

Handwritten examples illustrating the Unit Digit Theorem:

$$(12\underline{6})^{239} \times (12\underline{2})^{1259}$$

6 × 8

8

QUESTION-1



#Q. $\frac{55^{55}}{11^{22}} + \frac{49^{34}}{7^6} + \frac{169^{52}}{13^{96}}$, find the unit digit of this expression

$$55 = 5 \times 11 \quad | \quad 169 = 13^2$$

$$49 = 7^2$$

5	55
11	11
	1

$$\frac{(5 \times 11)^{55}}{11^{22}} + \frac{(7^2)^{34}}{7^6} + \frac{(13^2)^{52}}{13^{96}}$$

$$\frac{5^{55} \times 11^{55}}{11^{22}} + \frac{7^{68}}{7^6} + \frac{13^{104}}{13^{96}}$$

$$(5^{55} \times 11^{33}) + 7^{62} + 13^8$$

$$(5 \times 1) + 9 + 1$$

$$5 + 9 + 1 = 5$$

A 5

B 7

C 9

D 3

	1	2	3	0
5	5	5	5	5
1	1	1	1	1
→ 7	7	9	3	1
→ 3	3	9	7	1

QUESTION- 2



#Q. Find the unit digit of $[87^{125!} \times 303^{198!}]$

$$(87)^{125!} \times (303)^{198!}$$

$$\underline{1} \times \underline{1} = 1$$

$$3! = 3 + 2 + 1$$

$$4! = 4 + 3 + 2 + 1$$

$$7! = 7 + 6 + 5 + 4 + 3 + 2 + 1$$

A 7

B 1 ✓

C 3

D None of these

	1	2	3	↓ 8
7	7	9	3	1
3	3	9	7	(1)

QUESTION- 3



#Q. 'abcd' is a 4 digit number formed by,

unit place digit of $3^x = \textcircled{a}$

unit place digit of $2^z = c$

unit place digit of $\textcircled{7^y} = b$

unit place digit of $8^t = d$

Where, x, y, z and t are natural number. Find the largest number 'abcd', such that each digit is distinct

$$a \rightarrow 3, \textcircled{9}, 7, 1$$

$$b \rightarrow \textcircled{7}, \cancel{9}, 3, 1$$

$$c \rightarrow \textcircled{2}, \textcircled{4}, \textcircled{8}, \textcircled{6}$$

$$d \rightarrow \textcircled{8}, \textcircled{4}, \textcircled{2}, \textcircled{6}$$

(Largest)
a b c d

→ 9786

$$(8)^t \quad (2)^z \quad (\underline{\underline{3}})^x \quad (7)^y$$

A 9876
Thousand
Hundred
Tens
Unit

B 9678

C 9786

D None of these

3	3	9	7	1
7	7	9	3	1
2	2	4	8	6
8	8	4	2	6

QUESTION- 4



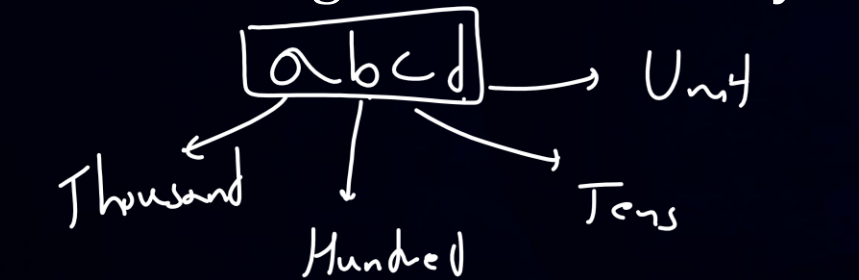
#Q. For a 4-digit number, the sum of its digits in the thousands, hundreds and tens places is 14, the sum of its digits in the hundreds, tens and units places is 15, and the tens place digit is 4 more than the units place digit. Then the highest possible 4-digit number satisfying the above conditions is

[CAT 2021 : Slot 2]

4195

Number ↑ a ↑ d ↑

a b c d
4 1 9 5



$$a + b + c = 14 \quad | \quad b + c + d = 15$$

$$d - a = 1$$

$$d - 1 = a$$

$$5 - 1 = a$$

$$4 = a$$

	$c = d + 4$	
	d	c
H	9	13
H	8	12
H	7	11
K	6	10
K	5	9



Number & Sum of Factors :



$$N = 48$$

$$\begin{aligned} 48 &= 1 \times 48 \\ &= 2 \times 24 \\ &= 3 \times 16 \\ &= 4 \times 12 \\ &= 6 \times 8 \end{aligned}$$

No. of factors = 10

$$\begin{aligned} \text{Sum} &= 1 + 48 + 2 + 24 + 3 + 16 \\ &\quad + 4 + 12 + 6 + 8 \\ &= 124 \end{aligned}$$

$$N = 60$$

$$\begin{aligned} 60 &= 1 \times 60 \\ &= 2 \times 30 \\ &= 3 \times 20 \\ &= 4 \times 15 \\ &= 5 \times 12 \\ &= 6 \times 10 \end{aligned}$$

No. of factors = 12

$$\begin{aligned} \text{Sum of factors} &= 1 + 60 + 2 + 30 + 3 + 20 + 4 + 15 + 5 + 12 + 6 + 10 \\ &= 168 \end{aligned}$$

$$\begin{array}{r|l} 2 & 60 \\ \hline 2 & 30 \\ \hline 3 & 15 \\ \hline 5 & 5 \\ \hline & 1 \end{array}$$

$$60 = 2^2 \times 3^1 \times 5^1$$

$$\begin{aligned} \text{Number} &= (2+1)(1+1)(1+1) \\ &= 3 \times 2 \times 2 \\ &= 12 \end{aligned}$$

$$\begin{aligned} \text{Sum} &= (2^0 + 2^1 + 2^2)(3^0 + 3^1)(5^0 + 5^1) \\ &= (1 + 2 + 4)(1 + 3)(1 + 5) \\ &= 7 \times 4 \times 6 \\ &= 168 \end{aligned}$$



Sum & Number of Factors (Divisors) :



(Case 1):

$$64800 = \underline{2}^{\textcircled{5}} \times \underline{3}^{\textcircled{4}} \times \underline{5}^{\textcircled{2}}$$

Total Number of all the Factors (Divisors) of 64800 :

$$\begin{aligned} &= (\underline{5+1}) \cdot (\underline{4+1}) \cdot (\underline{2+1}) \\ &= 6 \times 5 \times 3 = \underline{90} \checkmark \end{aligned}$$

Sum of all the Factors (Divisors) of 64800 :

$$\begin{aligned} &= (2^0 + 2^1 + 2^2 + 2^3 + 2^4 + 2^5) \cdot (3^0 + 3^1 + 3^2 + 3^3 + 3^4) \cdot (5^0 + 5^1 + 5^2) \\ &= (1 + 2 + 4 + 8 + 16 + 32) \cdot (1 + 3 + 9 + 27 + 81) \cdot (1 + 5 + 25) \\ &= 63 \times 121 \times 31 = \boxed{236313} \end{aligned}$$

$$\begin{array}{r} 2 \overline{) 64800} \\ \underline{2} 32400 \\ 2 \overline{) 32400} \\ \underline{2} 16200 \\ 2 \overline{) 16200} \\ \underline{2} 8100 \\ 2 \overline{) 8100} \\ \underline{2} 4050 \\ 2 \overline{) 4050} \\ \underline{3} 2025 \\ 3 \overline{) 2025} \\ \underline{3} 675 \\ 3 \overline{) 675} \\ \underline{3} 225 \\ 3 \overline{) 225} \\ \underline{3} 75 \\ 3 \overline{) 75} \\ \underline{5} 25 \\ 5 \overline{) 25} \\ \underline{5} 5 \\ 5 \overline{) 5} \\ \underline{5} 0 \end{array}$$



Sum & Number of Factors (Divisors) :



$$64800 = 2^5 \times 3^4 \times 5^2$$

(Case 2) :

Total Number of all Odd Factors (Divisors) of 64800 :

$$= \binom{1}{1} \cdot (4+1)(2+1)$$

$$= (1)(5)(3) = \underline{\underline{15}}$$

Sum of all Odd Factors (Divisors) of 64800 :

$$2^0 = 1$$

$$2^1 = 2$$

$$2^2 = 4$$

$$2^3 = 8$$

$$2^4 = 16$$

$$\vdots$$

$$= \binom{2^0}{2^0} \cdot (3^0 + 3^1 + 3^2 + 3^3 + 3^4) (5^0 + 5^1 + 5^2)$$

$$= (1)(1+3+9+27+81)(1+5+25)$$

$$= 1 \times 121 \times 31 = 3751$$

60

$$= \boxed{\underline{1}, 60, \underline{2}, 30, \underline{3}, 20, 4, \underline{15}, \underline{5}, 12, 8, 10}$$

odd $\rightarrow (4)$

$$\text{Sum} = 1 + 3 + 15 + 5 = 24$$

$$60 = 2^2 \times 3^1 \times 5^1$$

$$\text{No.} = (1)(1+1)(1+1) = 2 \times 2 = 4$$

$$\text{Sum} = \binom{2^0}{2^0} (3^0 + 3^1) (5^0 + 5^1) \\ = (1)(1+3)(1+5) = 24$$



Sum & Number of Factors (Divisors) :

(Case 3) :

$$64800 = 2^5 \times 3^4 \times 5^2$$

Total Number of all Even Factors (Divisors) of 64800 :

power

$$\begin{aligned} &= (5)(4+1)(2+1) \\ &= 5 \times 5 \times 3 = 75 \end{aligned}$$

Sum of all Even Factors (Divisors) of 64800 :

$$\begin{aligned} &= (2^1 + 2^2 + 2^3 + 2^4 + 2^5)(3^0 + 3^1 + 3^2 + 3^3 + 3^4)(5^0 + 5^1 + 5^2) \\ &= (2 + 4 + 8 + 16 + 32)(1 + 3 + 9 + 27 + 81)(1 + 5 + 25) \\ &= (62)(121)(31) \\ &= \underline{\underline{232562}} \end{aligned}$$



Sum & Number of Factors (Divisors) :

(Case 4) : $\frac{5}{2} = 2 \dots 64800 = 2^5 \times 3^4 \times 5^2$ $\frac{5}{2}$ $\frac{2}{2}$

Total Number of all Perfect Square Factors (Divisors) of 64800 :

$$= (2 + 1)(2 + 1)(1 + 1)$$

$$= 3 \times 3 \times 2 = \underline{\underline{18}}$$

Sum of all Perfect Square Factors (Divisors) of 64800 :

$$= (2^0 + 2^2 + 2^4)(3^0 + 3^2 + 3^4)(5^0 + 5^2)$$

$$= (1 + 4 + 16)(1 + 9 + 81)(1 + 25)$$

$$= 21 \times 91 \times 26 = 49186$$



Sum & Number of Factors (Divisors) :

(Case 5) :

Total Number of all Perfect Cube Factors (Divisors) of 64800 :

$$\begin{aligned}
 &= (1 + 1)(1 + 1)(0 + 1) \\
 &= 2 \times 2 \times 1 = 4
 \end{aligned}$$

Sum of all Perfect Cube Factors (Divisors) of 64800 :

$$\begin{aligned}
 &= (2^0 + 2^3)(3^0 + 3^3)(5^0) \\
 &= (1 + 8)(1 + 27)(1) \\
 &= 9 \times 28 \times 1 = 252
 \end{aligned}$$

60
↓
1, 60, 2, 30,
3, 20, 4, 15,
5, 12, 6, 10

$$\frac{5}{3} = 1 \dots$$

$$64800 = 2^5 \times 3^4 \times 5^2$$

$$\frac{4}{3} = 1 \dots$$

$$\frac{2}{3} = 0 \dots$$

QUESTION- 6



#Q. How many factors of 216576 are divisible by 144?

TRY

(Next class)

A 14

B 12

C 16

D 18

$$64800 = \underline{2}^5 \times \underline{3}^4 \times \underline{5}^2$$

No. of prime factors = 3

$$\text{Sum} = 2 + 3 + 5 = 10$$



2 Mins Summary

- 1) Concepts & Numerical based on 'Number System Basics'
- 2) Concepts based on 'Unit Digit'
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- 4) Problems for Practice & CAT PYQs



*Thank
You*